

GIT & GITHUB BASICS

PARIKSHIT MIT — TASK 1

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1. INTRODUCTION

Git and GitHub are important tools used for managing and collaborating on software and technical projects. GitHub is an online platform that hosts Git repositories and allows people to store, share, and collaborate on projects.

This report provides an introduction to Git and GitHub and explains basic concepts such as repositories, commits, branches, push, pull, and clone. It also documents the practical use of GitHub for completing Task 1.

2. WHAT IS GIT?

Git is a distributed version control system used to track changes in files and projects. Git is especially useful when working on projects where files are changed frequently or when multiple people are working on the same project. It helps maintain a history of changes and makes collaboration easier.

For example, while developing a website for a project, Git can record changes made to HTML and CSS files. If a mistake is made later, previous versions can be reviewed or restored.

4. GIT VS GITHUB

Git is a version control system that runs on a computer and tracks changes made to project files. GitHub is an online platform that hosts Git repositories and provides tools for collaboration and project management.

The main difference is that Git is the tool used to track versions, while GitHub is an online service used to store and share those Git-based projects.

5. REPOSITORY

A repository, commonly called a repo, is a storage location for a project and its files. A repository can contain source code, documents, images, and other project-related files.

For this task, a GitHub repository named PARIKSHIT-1 was created to organize and store the work related to Task 1.

6. COMMIT

A commit is a saved record of changes made to files in a Git repository. Each commit contains information about the changes and usually has a short message describing what was changed.

Commits help maintain a history of the project and make it possible to understand what changes were made over time.

7. BRANCH

A branch is a separate line of development in a Git repository. It allows changes to be made without directly affecting the main project.

The main branch is commonly called "main". For this task, the work is being maintained on the main branch.

8. PUSH

Push is the process of sending committed changes from a local computer to a remote GitHub repository.

Computer → GitHub = Push

9. PULL

Pull is the process of receiving the latest changes from a remote GitHub repository to a local computer.

GitHub → Computer = Pull

10. CLONE

Clone means creating a copy of a GitHub repository on a local computer. It allows a user to work on the project locally.

A PARIKSHIT team member can clone a repository, make changes on their computer, and later push the changes back to GitHub.

11. BASIC GIT WORKFLOW

A basic Git workflow consists of making changes, checking those changes, saving them as commits, and sharing them with a remote repository such as GitHub.

1. Create or modify project files.
2. Check the changes.
3. Stage the changes.
4. Commit the changes with a meaningful message.
5. Push the commits to GitHub.
6. Other team members can pull the latest changes

12. PRACTICAL WORK COMPLETED

As part of this task, a GitHub account was created and a repository named PARIKSHIT-1 was created for the Task 1 activities.

A README.md file was added to the repository. The README file was edited to include information about the project, the assigned subsystem, and the tasks included in Task 1.

The changes were saved using a GitHub commit with the message "Update README for Task 1". This provided practical experience with creating a repository, editing files, and committing changes using GitHub.

13. KEY LEARNINGS

Through this task, I learned the basic purpose and use of Git and GitHub. I understood how repositories are used to organize projects and how commits help maintain a history of changes.

I also learned the basic concepts of branches, push, pull, and clone. Most importantly, I gained practical experience by creating and managing a GitHub repository for the PARIKSHIT MIT Task 1.

14. CONCLUSION

Git and GitHub are useful tools for managing projects, tracking changes, and collaborating with team members. Understanding these basic concepts is important for working on technical projects where multiple people contribute to the same work.

The practical use of GitHub during this task provided a foundation for maintaining and sharing project work in an organized manner. These skills will also be useful for future technical and software-related activities in the PARIKSHIT MIT Student Satellite Project.